Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	Waves spread out [at each slit] (1) Overlap of [diffracted] waves allows <u>interference / superposition</u> (1)	2			2		
		(ii)	$\lambda = \frac{0.45\times10^{-3}\times2.2\times10^{-3}}{1.8}$ [Tolerate wrong powers of 10] (1) λ = 550 n[m] (1)	1	1		2	2	
		(iii)	[For any bright fringe], path difference must be $n\lambda$ $[n = 0, 1, 2 \ldots]$ (1) can be implied For P, $n = 3$ **or** path difference $= 3\lambda$ or by implication (1) So path difference = 1650 n[m] **ecf** on value of λ from (a)(ii) (1) (Geometrical method that does not use λ gains maximum of 1)	1	1 1		3		
	(b)		Orders / bright fringes much further apart [for grating] (1) Orders / bright fringes much sharper (1) Accept brighter or increased intensity. Don't accept clearer	2			2		2
			Question 5 total	**6**	**3**	**0**	**9**	**0**	**2**